Research Notes on Layer normalization

Layer normalization

>> Section 1

x i ^ = x i − μ σ 2 + ■ , y i = γ i x i ^ + β i {\displaystyle {\hat {x_{i}}}={\frac {x_{i}-\mu }{\sqrt {\sigma ^{2}+\epsilon }}},\quad y_{i}=\gamma _{i}{\hat {x_{i}}}+\beta _{i}}

>> Section 2

x h , w , c (l) = ∑ h ′ , w ′ , c ′ K h ′ − h , w ′ − w , c , c ′ (l) x h ′ , w ′ , c ′ (l − 1) + b c (l) {\displaystyle x_{h,w,c}^{(l)}=\sum _{h',w',c'}K_{h'-h,w'-w,c,c'}^{(l)}x_{h',w',c'}^{(l-1)}+b_{c}^{(l)}}

>> Section 3

x i ^ = x i 1 D ∑ j = 1 D x j 2 , y i = γ x i ^ + β {\displaystyle {\hat {x_{i}}}={\frac {x_{i}}{\sqrt {{\frac {1}{D}}\sum _{j=1}^{D}x_{j}^{2}}}},\quad y_{i}=\gamma {\hat {x_{i}}}+\beta }

>> Section 4

■ ■ x (l) ■ B N (x (l)) ■ x (l + 1) ■ ■ {\displaystyle \cdots \mapsto x^{(l)}\mapsto \mathrm {BN} (x^{(l)})\mapsto x^{(l+1)}\mapsto \cdots }

>> Section 5

Iterate x ■ 1 ■ W x ■ 2 W x {\displaystyle x\mapsto {\frac {1}{\|Wx\|_{2}}}Wx} to convergence x ∗ {\displaystyle x^{*}} . This is the eigenvector of W {\displaystyle W} with eigenvalue ■ W ■ s {\displaystyle \|W\|_{s}} .

>> Section 6

μ (t) = 1 D ∑ i = 1 D x i (t) , (σ (t)) 2 = 1 D ∑ i = 1 D (x i (t) − μ (t)) 2 {\displaystyle \mu ^{(t)}={\frac {1}{D}}\sum _{i=1}^{D}x_{i}^{(t)},\quad (\sigma ^{(t)})^{2}={\frac {1}{D}}\sum _{i=1}^{D}(x_{i}^{(t)}-\mu ^{(t)})^{2}}

>> Section 7

where each network module can be a linear transform, a nonlinear activation function, a convolution, etc. x (0) {\displaystyle x^{(0)}} is the input vector, x (1) {\displaystyle x^{(1)}} is the output vector from the first module, etc. BatchNorm is a module that can be inserted at any point in the feedforward network. For example, suppose it is inserted just after x (l) {\displaystyle x^{(l)}} , then the network would operate accordingly:

>> Section 8

μ c (l) = 1 B H W ∑ b = 1 B ∑ h = 1 H ∑ w = 1 W x (b) , h , w , c (l) (σ c (l)) 2 = 1 B H W ∑ b = 1 B ∑ h = 1 H ∑ w = 1 W (x (b) , h , w , c (l) − μ c (l)) 2 {\displaystyle {\begin{aligned}\mu _{c}^{(l)}&={\frac {1}{BHW}}\sum _{b=1}^{B}\sum _{h=1}^{H}\sum _{w=1}^{W}x_{(b),h,w,c}^{(l)}\\(\sigma _{c}^{(l)})^{2}&={\frac {1}{BHW}}\sum _{b=1}^{B}\sum _{h=1}^{H}\sum _{w=1}^{W}(x_{(b),h,w,c}^{(l)}-\mu _{c}^{(l)})^{2}\end{aligned}}}

>> Section 9

Group normalization Group normalization (GroupNorm) is a technique also solely used for CNNs. It can be understood as the LayerNorm for CNN applied once per channel group. Suppose at a layer l {\displaystyle l} , there are channels 1 , 2 , … , C {\displaystyle 1,2,\dots ,C} , then it is partitioned into groups g 1 , g 2 , … , g G {\displaystyle g_{1},g_{2},\dots ,g_{G}} . Then, LayerNorm is applied to each group.

>> Section 10

μ = α E [x] + (1 − α) μ x , train σ 2 = (α E [x] 2 + (1 − α) μ x 2 , train) − μ 2 {\displaystyle {\begin{aligned}\mu &=\alpha E[x]+(1-\alpha)\mu _{x,{\text{ train}}}\\\sigma ^{2}&=(\alpha E[x]^{2}+(1-\alpha)\mu _{x^{2},{\text{ train}}})-\mu ^{2}\end{aligned}}}

>> Section 11

y h , w , c (l) , content = σ c (l) , style (x h , w , c (l) , content − μ c (l) , content (σ c (l) , content) 2 + ■) + μ c (l) , style {\displaystyle {\begin{aligned}y_{h,w,c}^{(l),{\text{ content}}}&=\sigma _{c}^{(l),{\text{ style}}}\left({\frac {x_{h,w,c}^{(l),{\text{ content}}}-\mu _{c}^{(l),{\text{ content}}}}{\sqrt {(\sigma _{c}^{(l),{\text{ content}}})^{2}+\epsilon }}}\right)+\mu _{c}^{(l),{\text{ style}}}\end{aligned}}}

>> Section 12

x ^ (b) , h , w , c (l) = x (b) , h , w , c (l) − μ c (l) (σ c (l)) 2 + ■ y (b) , h , w , c (l) = γ c x ^ (b) , h , w , c (l) + β c {\displaystyle {\begin{aligned}{\hat {x}}_{(b),h,w,c}^{(l)}&={\frac {x_{(b),h,w,c}^{(l)}-\mu _{c}^{(l)}}{\sqrt {(\sigma _{c}^{(l)})^{2}+\epsilon }}}\\y_{(b),h,w,c}^{(l)}&=\gamma _{c}{\hat {x}}_{(b),h,w,c}^{(l)}+\beta _{c}\end{aligned}}}

>> Section 13

where i {\displaystyle i} indexes the coordinates of the vectors, and b {\displaystyle b} indexes the elements of the batch. In other words, we are considering the i {\displaystyle i} -th coordinate of each vector in the batch, and computing the mean and variance of these numbers. It then normalizes each coordinate to have zero mean and unit variance:

>> Section 14

increase the speed of training convergence, reduce sensitivity to variations and feature scales in input data, reduce overfitting, and produce better model generalization to unseen data. Normalization techniques are often theoretically justified as reducing covariance shift, smoothing optimization landscapes, and increasing regularization, though they are mainly justified by empirical success.

>> Section 15

ghost batching: randomly partition a batch into sub-batches and perform BatchNorm separately on each; weight decay on γ {\displaystyle \gamma } and β {\displaystyle \beta } ; and combining BatchNorm with GroupNorm. A particular problem with BatchNorm is that during training, the mean and variance are calculated on the fly for each batch (usually as an exponential moving average), but during inference, the mean and variance were frozen from those calculated during training. This train-test disparity degrades performance. The disparity can be decreased by simulating the moving average during inference:

>> Section 16

where a x , y i {\displaystyle a_{x,y}^{i}} is the activation of the neuron at location (x , y) {\displaystyle (x,y)} and channel i {\displaystyle i} . I.e., each pixel in a channel is suppressed by the activations of the same pixel in its adjacent channels.

>> Section 17

b c (l) {\displaystyle b_{c}^{(l)}} is the bias term for the c {\displaystyle c} -th channel of the l {\displaystyle l} -th layer. In order to preserve the translational invariance, BatchNorm treats all outputs from the same kernel in the same batch as more data in a batch. That is, it is applied once per kernel c {\displaystyle c} (equivalently, once per channel c {\displaystyle c}), not per activation x h , w , c (l + 1) {\displaystyle x_{h,w,c}^{(l+1)}} :